

Foundations & Basics

Database Design & Modeling



By: Zherou A. Majied

Conceptual vs Logical vs Physical Design

- Conceptual Design

What data do we need? (High-level business view)



- Logical Design

How data is structured? (Relational rules)



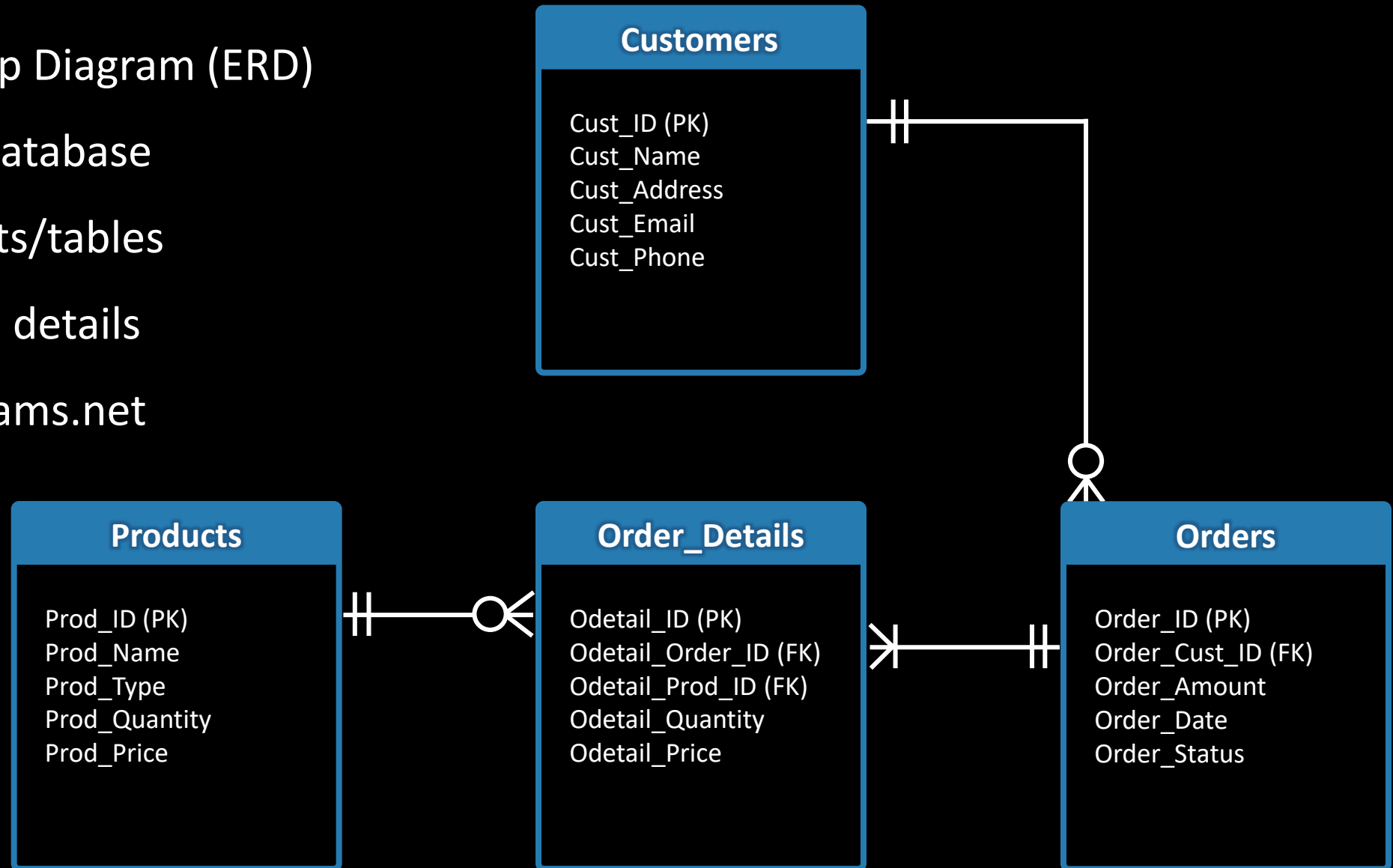
- Physical Design

How data is implemented? (Technical details)



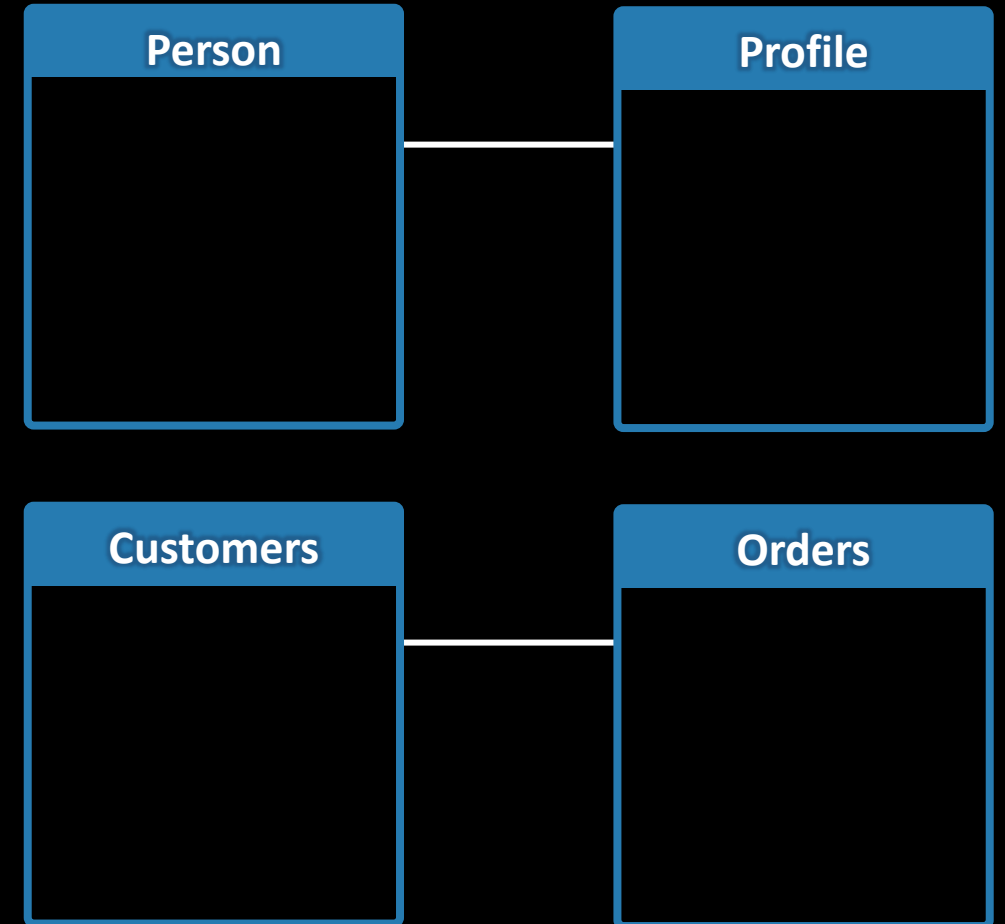
What is an ER Model?

- Entity Relationship Diagram (ERD)
- Blueprint of the database
- Entities are objects/tables
- Attributes are the details
- <https://app.diagrams.net>



Relationships & Cardinality in ERD

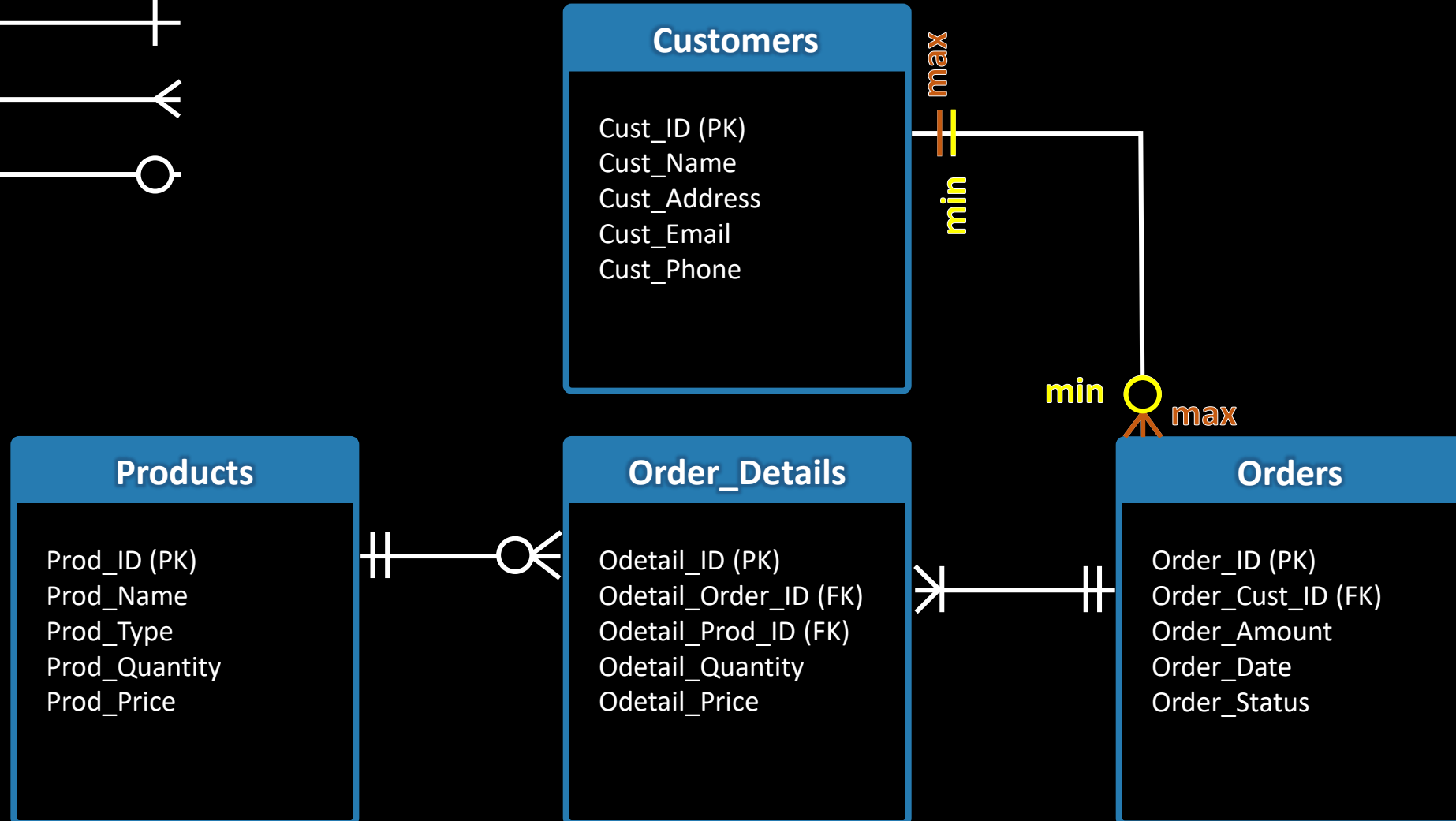
- Relationship: Link between entities
 - One-to-One
 - One-to-Many
 - Many-to-Many



Relationships & Cardinality in ERD

- Cardinality: Defines the link in numerical context

- One 
- Many 
- Zero (Optional) 



Core Keys in Database Design

- Primary Key (**PK**): Uniquely identifies a record
- Foreign Key (**FK**): Links records between tables

Odetail_ID	Odetail_Order_ID	Odetail_Prod_ID	Odetail_Quantity	Odetail_Price
605	10001	90	5	5.00
606	10001	1058	1	17.00
...	...	...	...	...
998	10120	91	2	13.00

- Candidate Key: Any column that could be a PK
- Composite Key: PK made of 2+ columns
- Alternate Key: Candidate Key not chosen as PK

Advanced Keys in Database Design

- **Super Key**

Any set of columns that uniquely identifies a record

- **Surrogate Key**

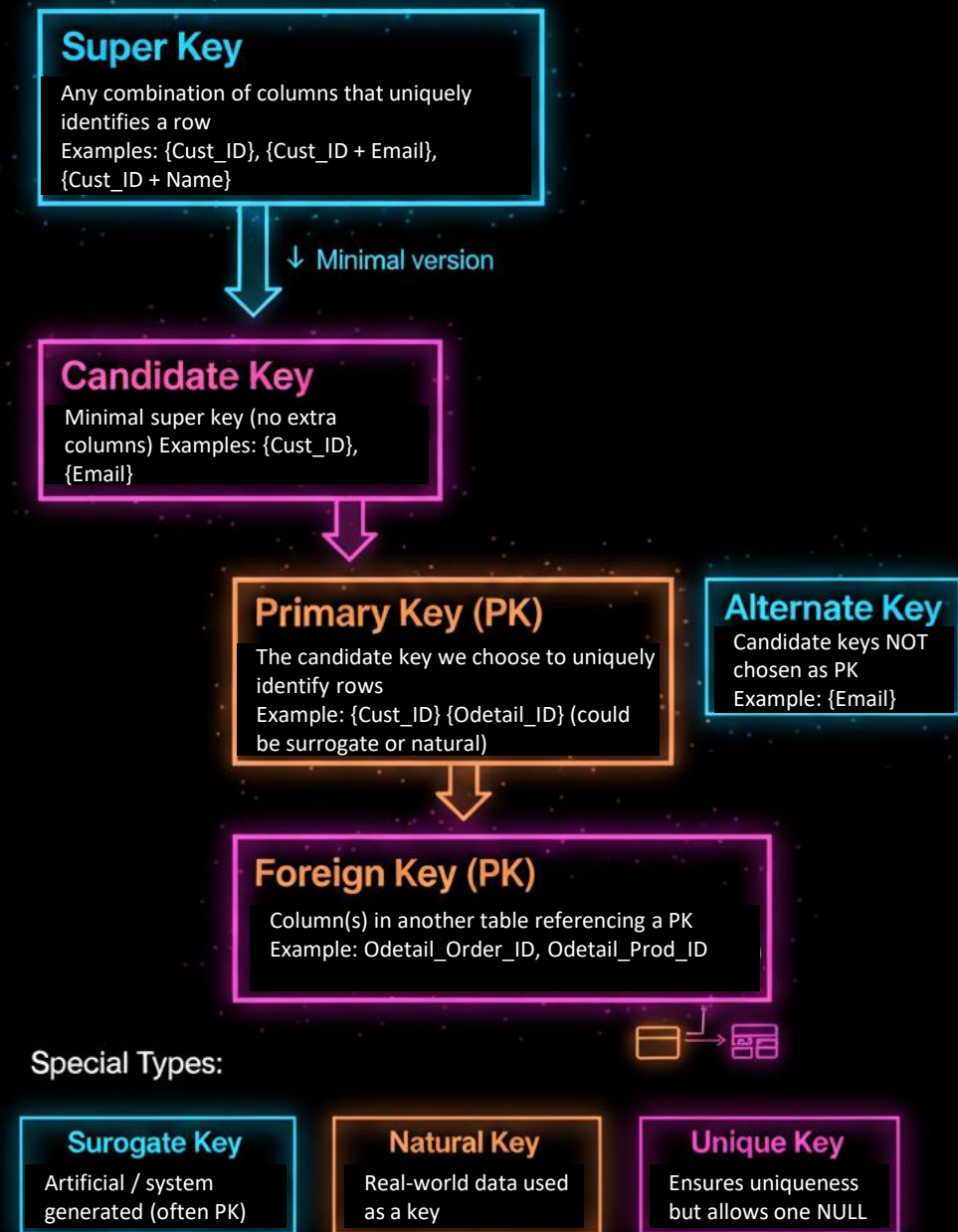
Artificial, system generated key

- **Natural Key**

Real-world data used as a key

- **Unique Key**

Ensures uniqueness but allows one NULL



Constraints in Database Design

- **NOT NULL:** Ensures that a column always has a value
- **DEFAULT:** Assigns a default value automatically
- **CHECK:** Validates data based on condition
- **UNIQUE:** Ensures that all values in a column are different
- **PK:** Uniquely identifies a record
- **FK:** Links two tables